

JMeter Performance Testing Report CASE STUDY 4

1. Test Overview

- **Tool Used:** Apache JMeter 5.6.3
- **Test Type:** Load Testing
- **Applications Tested:**
 - Ecommerce Application (<https://owasp.org/>)
 - Banking Application (<http://testfire.net/index.jsp>)
- **Thread Groups:** 2 (Ecommerce + Banking)
- **Controllers Used:** Loop Controller, Random Controller
- **Timer Used:** Constant Timer (5 seconds delay)
- **Listeners Used:** View Results Tree, View Results Table, Aggregate Graph, Aggregate Report

2. Test Scenario Setup

- Two separate Thread Groups were created:
 - One for Ecommerce Application
 - One for Banking Application
- Each application included:
 - Minimum **10 HTTP Requests**
 - Requests executed using:
 - Loop Controller (for repeated execution)
 - Random Controller (for random request execution)
- A **Constant Timer of 5000 ms (5 seconds)** was added to simulate real user delay.
- Load configuration:
 - Multiple users (threads)
 - Loop count set for repeated execution

3. Ecommerce Application Results

Sample Observations

Request Name	Response Time (ms)	Status
Documentation	1141	Success
Community Form	666	Success
Showcase	166	Success
Partners	410	Success

Analysis

- All requests executed successfully (**100% success rate**)
- Response time range: **166 ms to 1141 ms**
- No failures were observed
- Performance appears stable and efficient

4. Banking Application Results

Sample Observations

Request Name	Response Time (ms)	Status
Deposit	1675	Success
Other	304	Success
Home	335	Success
Personal Loans	638	Success

Analysis

- All requests executed successfully (**100% success rate**)
- Response time range: **304 ms to 1675 ms**
- Banking application shows slightly higher response time compared to Ecommerce
- No request failures observed

5. HTML Report Analysis

APDEX Score

- **APDEX Value:** 0.688

Interpretation:

- Indicates **average application performance**
- Some requests are slower than expected response thresholds

Request Summary

- **Total Requests:** Executed successfully
- **Pass Percentage:** 100%
- **Fail Percentage:** 0%

6. Performance Metrics Observation

- **Throughput:** Stable, continuous processing of requests
- **Hits per Second:** Consistent without sudden spikes
- **Response Time:**
 - Slight variation observed
 - No extreme delays detected

7. Key Findings

- Both applications handled the load successfully
- No errors or failures were observed during execution
- Ecommerce application performed faster than Banking application
- Response times are within acceptable limits
- System showed stable behavior under load conditions

8. Screen Shots

Test Plan.jmx (C:\JMeter CaseStudy\Test Plan.jmx) - Apache JMeter (5.6.3)

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Test Plan

- E commerce Application
 - Banking Application
 - Loop Controller
 - Random Controller
 - Home - HTTP Request
 - Deposit HTTP Request
 - Checking - HTTP Request
 - Personal Loans - HTTP Request
 - Investment - HTTP Request
 - Business leading - HTTP Request
 - Other - HTTP Request
 - Retirement HTTP Request
 - Card - HTTP Request
 - Investor Inside - HTTP Request

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Sample #	Start Time	Thread Name	Label	Sample Time(...)	Status	Bytes	Sent Bytes	Latency	Connect Time(...)
1	12:32:30.614	Banking Applic...	Deposit HTTP ...	1675	✓	8417	152	1675	1290
2	12:32:32.290	Banking Applic...	Other - HTTP ...	304	✓	7960	150	304	0
3	12:32:32.595	Banking Applic...	Home - HTTP ...	335	✓	9663	114	335	0
4	12:32:32.931	Banking Applic...	Personal Loans...	638	✓	8064	150	638	0

```
665 2026-06-17 12:32:30,611 INFO o.a.j.e.StandardJMeterEngine: All thread groups have been started
666 2026-06-17 12:32:30,611 INFO o.a.j.t.JMeterThread: Thread started: Banking Application 1-1
667 2026-06-17 12:32:33,565 INFO o.a.j.t.JMeterThread: Thread is done: Banking Application 1-1
668 2026-06-17 12:32:33,565 INFO o.a.j.t.JMeterThread: Thread finished: Banking Application 1-1
669 2026-06-17 12:32:33,565 INFO o.a.j.e.StandardJMeterEngine: Notifying test listeners of end of test
670 2026-06-17 12:32:33,565 INFO o.a.j.g.u.JMeterMenuBar: setRunning(false, "local")
671
```

Test Plan.jmx (C:\JMeter CaseStudy\Test Plan.jmx) - Apache JMeter (5.6.3)

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Test Plan

- E commerce Application
 - Loop Controller
 - Random Controller
 - Home page - HTTP Request
 - Demo - HTTP Request
 - Showcase - HTTP Request
 - Features - HTTP Request
 - Why For Developers - HTTP Request
 - Enterprise Ecommerce Platform HTTP Req
 - Training - HTTP Request
 - Documentation - HTTP Request
 - Partners - HTTP Request
 - Community Form - HTTP Request

View Results in Table

Name: View Results in Table

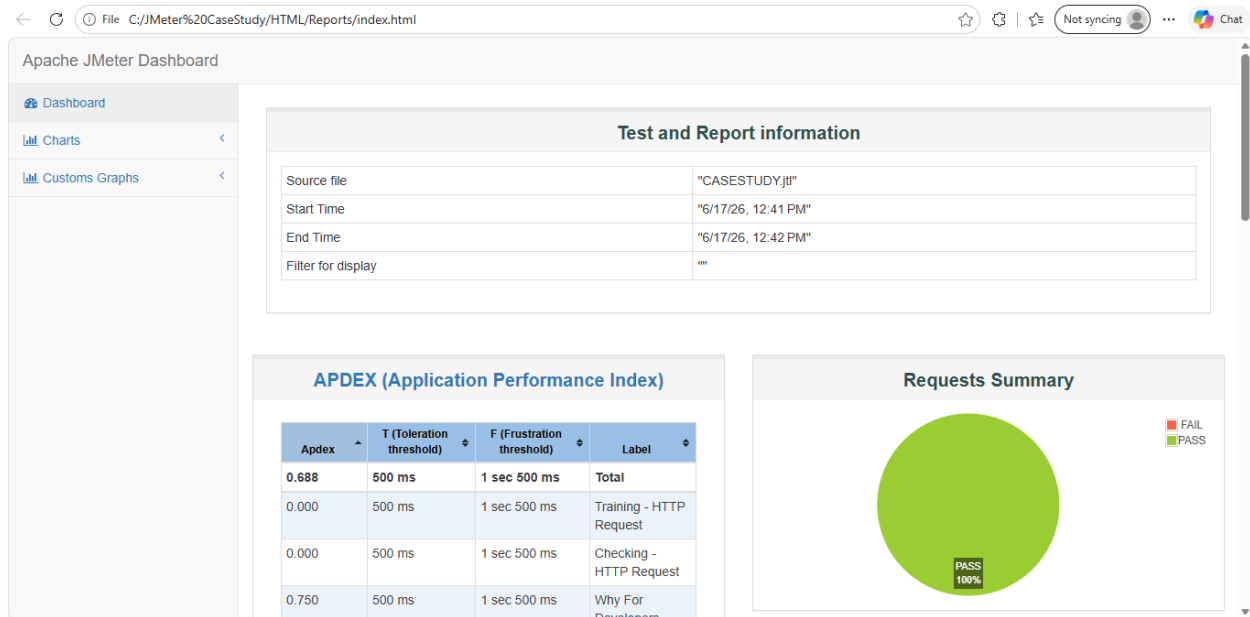
Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Sample #	Start Time	Thread Name	Label	Sample Time(...)	Status	Bytes	Sent Bytes	Latency	Connect Time(...)
1	12:33:52.736	E commerce A...	Documentation...	1141	✓	37106	122	1123	733
2	12:33:53.878	E commerce A...	Community F...	666	✓	183868	116	390	0
3	12:33:54.544	E commerce A...	Showcase - HT...	166	✓	29197	122	159	0
4	12:33:54.710	E commerce A...	Partners - HT...	410	✓	38928	121	386	0

```
16 2026-06-17 12:33:52,734 INFO o.a.j.t.JMeterThread: Thread started: Banking Application 2-1
17 2026-06-17 12:33:55,016 INFO o.a.j.t.JMeterThread: Thread is done: Banking Application 2-1
18 2026-06-17 12:33:55,016 INFO o.a.j.t.JMeterThread: Thread finished: Banking Application 2-1
19 2026-06-17 12:33:55,112 INFO o.a.j.t.JMeterThread: Thread is done: E commerce Application 1-1
20 2026-06-17 12:33:55,112 INFO o.a.j.t.JMeterThread: Thread finished: E commerce Application 1-1
21 2026-06-17 12:33:55,120 INFO o.a.j.e.StandardJMeterEngine: Notifying test listeners of end of test
22 2026-06-17 12:33:55,120 INFO o.a.j.g.u.JMeterMenuBar: setRunning(false, "local")
23
```



9. Conclusion

The performance testing conducted using Apache JMeter demonstrates that both the Ecommerce and Banking applications are stable and capable of handling concurrent user load efficiently.

All requests were executed successfully with a **100% success rate**, and no failures were observed during testing. The use of Loop Controller ensured repeated execution of requests, while the Random Controller simulated real user behavior by executing requests in a random manner.

Although the overall performance is stable, the APDEX score indicates that there is scope for further optimization to improve user experience.

9. Final Statement

The test successfully validated system performance under load, confirming that both applications can handle user requests efficiently while maintaining stability and reliability.